

STRATA: A Stratified Process-Mining Framework for HVAC Fault Detection and Diagnosis

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Abstract

Faults in building HVAC equipment persist because the tools meant to catch them either raise alarms that operators learn to ignore, or flag anomalies that nobody can act on. STRATA turns ordinary one-minute trend data into an event log along the control sequence, discovers process models at the device and unit level, and scores each day through eight calibrated, physically interpretable channels against a measured noise floor, so that every alarm names a location, states its evidence and is held to a false-alarm budget. It was built under a falsification protocol: every hypothesis was pre-registered with the observation that would refute it, every result was recomputed independently before it was quoted, and automated tests pin the published numbers and the falsifiers that fired.

On five public LBNL benchmark systems, including the two fan-powered terminal units for which no detection results had been published before, STRATA detects 158 of 175 seeded faults, most on the first day, at 26 false-alarm days in 480 held-out fault-free days (1 to 9.4% per system). Two systems were onboarded blind, by configuration alone. The alarm names the correct terminal unit in 37 of the 50 detections with known ground truth and a wrong one in none, and the faulted subsystem in 83% of all detections. The false-alarm budget holds under injected sensor error, change-of-value logging and schedule shifts (worst 7 of 96 days) and on one measured rooftop unit (3 of 49).

Process mining helps to get detection but is not detecting on its own. The strata and the event abstraction carry the localisation, and invariants that discovery surfaced became several of the rules that detect. Conformance checking of the discovered models, pre-registered as the detector, was refuted: over 199 scenarios on six systems, no fault is caught by the models alone. A principal-component baseline detects three fewer of the 73 scenarios it shares and cannot name the device. A third-party Guideline 36 rule battery, held to the same gate, matches STRATA within one detection on the air handler and detects a third as many faults on the terminal units, so the pre-registered improvement claim over current practice is withdrawn there. The evaluation also produced eight machine-verified errata in the benchmark datasets. Every number regenerates from the public repository.

1 Introduction

Buildings and construction account for 32% of global final energy use and 34% of energy-related carbon dioxide emissions [57]. HVAC accounts for up to half of an individual building’s consumption [7] and for about 30% of all commercial building energy in the United States [19], and 15–30% of that is wasted through faulty operation and poor control [7, 32]; correcting the common faults saves 5–30% of whole-building energy [7, 52]. The faults persist even where detection is installed. Across

60,000 pieces of equipment in 317 commercial buildings, Crowe and colleagues found 245 faults per building per month and 40% of air handlers in a faulted state on any given day, in buildings that run modern automation systems with alarms [12]. Seven of their ten most common faults are behaviour-based: wrong schedules, unreleased overrides, missed setpoint resets, broken economiser sequences. A threshold alarm or a pre-written rule [54] catches such a fault only if someone wrote that rule in advance [25]. The public benchmarks used in this paper seed component, sensor and control-loop faults and none of that class, so this argument motivates the design and is not tested by it.

Research has gone the other way. Machine learning has been surveyed across 211 HVAC fault-detection studies [7], with accuracies saturated at 97–99% on the standard benchmarks [51, 56] and interpretability added afterwards at the level of input features [11, 15, 36, 65, 67], which tells an operator that a sensor mattered, not what the building should have been doing. Of 65 primary studies audited in 2025, two shared code links, one of them broken, and 72% do not say whether their data is public [42].

What an operator needs is a detector that works from the trend data already recorded, names the unit and subsystem, explains the alarm in the physics of the equipment, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining, the one methodology built to model the sequence in which a system moves between states, has not been brought to this problem: to our knowledge no published work applies event-log process mining to building HVAC fault detection. The 211-study survey contains no process-mining entry [7], the reviews of process mining in Industry 4.0 and on sensor data contain no building HVAC application [2, 8], the Lawrence Berkeley National Laboratory taxonomy of fault-detection methods has no such leaf [24], and our own exact-phrase searches of the OpenAlex corpus return none [50].

STRATA applies process mining in stratified form: sensor streams are abstracted into events along ASHRAE Guideline 36 control sequences [3], Petri nets are discovered per device and per unit [43, 58], and the channels that read the log sit beside physics rules and residuals that read the raw signals, each calibrated on its own noise. Our first hypothesis, that the discovered net would itself be the detector, was pre-registered and refuted: a single hand-written rule out-covered the model channel where it was strongest. What settled is the division of labour, and it is a productive one. The strata and the event abstraction give every alarm its location, the frequency and oscillation channels are the only ones that catch the unstable-controller faults, invariants that discovery surfaced became several of the rules that detect, and conformance checking of the discovered nets, the part we expected to detect, contributes no detection of its own. Process mining helps to get detection but is not detecting on its own.

The paper contributes (1) the framework and the detector it yields, evaluated on five public benchmark systems including the two fan-powered terminal units for which no detection results had been published before; (2) what process mining contributes to HVAC fault detection, established under pre-registration: the first such application, whose structure gives the detector its location and several of its rules, and whose conformance channel catches no fault the calibrated channels miss; (3) an evaluation protocol of pre-registration, honoured falsifiers, independent recomputation and full artefact regeneration; and (4) eight machine-verified errata in the benchmark datasets.

2 Research Aim and Objectives

This research set out to build a fault detector for building HVAC systems that an operator can trust and act on: one that works from the trend data the building already records, explains each

alarm in the physics of the equipment, names the unit and the subsystem at fault, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining was the means chosen to get there, because it is the one methodology built to model the sequence in which a system moves between states, and no published work had applied it to this problem. The aim therefore carried a second question: what the discovered process models contribute to such a detector, and whether they detect on their own. Everything reported here is validated on public benchmark data; deployment on an operating building is the intended next phase, and Section ?? states what it would take.

A building operator today finds a fault by waiting for a complaint or by watching trend graphs for a curve that has drifted from where it usually sits; both find it late and neither names the equipment responsible. STRATA is meant to consume a building’s ordinary one-minute trend data and one configuration file naming its sensors and control rules, learn from a fault-free period what healthy operation looks like, and then, every day, name the days that break a physical relationship the healthy period established: which unit, which relationship, by how much, and how often that channel has cried wolf on healthy days. The output is a work order with a false-alarm budget the operator can hold the system to, and onboarding a new equipment type is a configuration exercise. That intent fixes six objectives, each reported with what was done and where it stands.

1. Abstract events from sensor data along Guideline 36 control sequences, with fault-signature events held out of model discovery. Each day is a case and every event is emitted by a configured rule: three kinds produce the state events model discovery may see, five produce fault-signature events it may not. The same code, with no building-specific additions, ran on seven LBNL datasets; the alphabet wall is enforced by unit tests and the circularity it prevents was measured. Met.
2. Discover models and establish what they contribute. Nets discovered at device, unit and cross-system strata on every system; their contribution measured under pre-registration on 199 scenarios across six systems. Met: the structure, the event log and the invariants they surfaced are what they contribute; the hypothesis that they would also detect by conformance alone is refuted.
3. Detect against a measured noise floor. Eight channels, every threshold from the fault-free year, every false-alarm rate out-of-sample; the budget held under injected field conditions and on one measured building. Met; the statistical repair that made this true cost seven detections, and that is stated.
4. Benchmark against known answers. 158 of 175 seeded scenarios detected at 26 false-alarm days in 480; against an audited PCA baseline, 64 of 73 to its 61; against the open-fdd Guideline 36 battery held to the same gate, within one detection on the air handler and three times the detections on the terminal units. All three parts executed; the improvement claim over current practice is withdrawn on the air handler.
5. Make every claim checkable. More than thirty pre-registered experiments, ten adverse results and four later fired falsifiers published, every headline number regenerated under a regression gate, a test suite that fails if a caveat is dropped. Met.
6. Release the work openly. Repository, configurations, artefacts, reproduction path and eight machine-verified dataset errata are public. Met.

What the objectives add up to is a detector that does what the intent asks on the benchmarks it was tested on, a measured and stated answer to the process-mining question, and a protocol that

made both checkable. What they do not yet add up to is a field result; Section ?? states what that would take. A hybrid with machine-learning classifiers was deferred until the models’ own detection had been established, and is future work.

3 Related Work

Machine learning for HVAC fault detection. Bi and colleagues survey 211 AI-based HVAC fault-detection studies from 2013 to 2023 [7]. Accuracies on the standard benchmarks have saturated: 98% across thirteen fault types with a random-forest and support-vector pipeline on RP-1312 data [56], 97% on three real buildings with a convolutional-and-LSTM hybrid whose authors name interpretability as the problem they did not solve [51]. The interpretable line pairs a deep model with a post-hoc explanation: Deng and colleagues’ graph network on the LBNL single-duct data reaches 88% at 500 labelled samples with per-sensor importance as the explanation [15]; attention and relevance propagation do the same on chiller data [11,36,65,67]. The explanation is at the level of input features, not of the control sequence the building was supposed to follow. The opportunity is a method whose model is its explanation.

Unsupervised methods and cross-building transfer. Real buildings rarely carry fault labels, so the strongest recent work is self-supervised: a masked-reconstruction Transformer on an unlabelled heat-pump building [1], and El Mokhtari and McArthur’s autoencoders across five fan coil units at Toronto Metropolitan University, the closest precedent for a Canadian university deployment [16]. Each yields a scalar anomaly score. Cross-building transfer in this literature means transferring learned weights by domain adaptation [17,18,35,69]; what STRATA transfers is a configuration file naming sensors and rules, a different operation, and the discovered models themselves did not transfer (Section ??).

Process mining and the empty quadrant. Process mining discovers models from event logs, checks conformance against them and enhances them with further perspectives [58,59]. Conformance was used directly as a trace-level anomaly detector [6,53] before neural sequence models overtook it on standard log corpora [45]; our own ablation, which finds no fault detected only by conformance, is consistent with that motivation. In adjacent cyber-physical domains the methodology has succeeded: robotic-arm fault diagnosis at an F1 of 98.9% [63], with alignment features feeding downstream classifiers for gains of 30 to 70 F1 points [64]; industrial IoT sensor validation [28]; industrial control systems [44,55]; and pasteurisation, where continuous sensor signals are abstracted into activity sequences and checked against a process specification [41]. The common barrier is event abstraction [9,13,39]. Building HVAC is an empty quadrant: the 211-study survey has no process-mining entry [7], the 47-study Industry 4.0 review [2] and the 36-study sensor-data review [8] contain no building HVAC application, construction-management process mining treats projects as cases and human tasks as activities [40], LBNL’s fault-detection taxonomy has no such leaf [24], and exact-phrase searches of OpenAlex [50] and the ICPM 2025 programme return none.

Rule-based practice. Fault detection in real buildings is rule-based: commercial platforms such as Automated Logic WebCTRL, installed at the Campus Activity Centre of Thompson Rivers University, ship libraries of over one hundred named faults, and Granderson and colleagues’ survey of fourteen commercial tools confirms the style [24,49]. LBNL’s automated fault-correction line applies hand-coded correction algorithms through BACnet writeback [37,38,52]. Both assume the faults of interest can be enumerated in advance. STRATA is complementary: it learns where

the building’s own healthy operation is invariant and holds those invariants at a measured false-alarm budget, and its located, evidenced alarms could in principle feed the correction algorithms. Section ?? measures both the strength and the limit of the rule library on the same benchmarks. Hand-designed Petri nets for HVAC [14, 20, 62, 68] are a different use of the formalism: authored by an engineer, with no event log and no discovery, and should not be conflated with the discovered models used here.

4 Method

STRATA (Figure 1; code and artefacts at <https://github.com/iyassh/strata>) consumes one fault-free year of ordinary building-management-system trend data and a single configuration file naming the building’s sensors and control rules. The configuration file is the only site-specific artefact; no code change was needed to onboard any system reported here. The framework is stratified in two senses: events are abstracted from the control sequence at device and unit granularity, so every alarm is born with a location, and the channels that read the resulting log (absence, frequency and the two conformance channels) sit beside physics rules, residuals and an oscillation channel that read the raw signals, each calibrated on its own noise.

Event abstraction and the alphabet wall. Continuous sensor streams become a discrete event log through control-sequence rules grounded in ASHRAE Guideline 36 [3], following Janssen and colleagues’ abstraction methodology [29] and the event-abstraction taxonomy of van Zelst and colleagues [61]. Each day is a case. Every event is emitted by a configured rule of one of eight kinds: occupancy, operating mode and time window produce state events, the only events admitted to model discovery; mismatch, leak, setpoint deviation, paired residual and envelope residual produce fault-signature events, which already encode engineering suspicion and are withheld from discovery. The partition, the alphabet wall, exists because of circularity, the process-mining form of data leakage [31, 66]: a model discovered over an alphabet that names faults will later “detect” them by recognising the project’s own rules. Measured on the single-duct system against the pre-partition prototype, the model channel’s apparent recall rises from 1.8% to 5.2% without the wall, while the deployed detector’s output is unchanged, because every leaked model detection was already a rules detection. The separation is enforced by unit tests: injecting signature events into a log cannot change a day’s fitness.

Discovery at three strata. Petri nets are discovered from the state-event log with the inductive miner [33, 34] in pm4py [5] at three granularities: the device stratum models an actuator’s lifecycle, the unit stratum the air handler’s daily cycle, and a cross-system grammar supports the transfer analysis of Section ?. Whether a net encodes order at all is checked directly by a reversal probe: every trace is reversed and re-scored. The single-duct net’s fitness falls from 0.886 to 0.553 under reversal; the two fan-powered nets do not move (0.953 and 0.902 to 0.903), so their fitness carries no ordering information.

Calibrated channels and the noise gate. Eight channels score each day independently (Table 1): rules from the APAR and Guideline 36 lineage [54] that name direct contradictions between measurements; residual bands on physical relationships that must hold in a known mode; model and device conformance, the alignment cost [10, 60] against the unit and device nets; absence of a scheduled activity; frequency and oscillation bands on activity counts and direction reversals; and a seasonal rate channel that corroborates but may not alarm alone. Calibration uses the fault-free

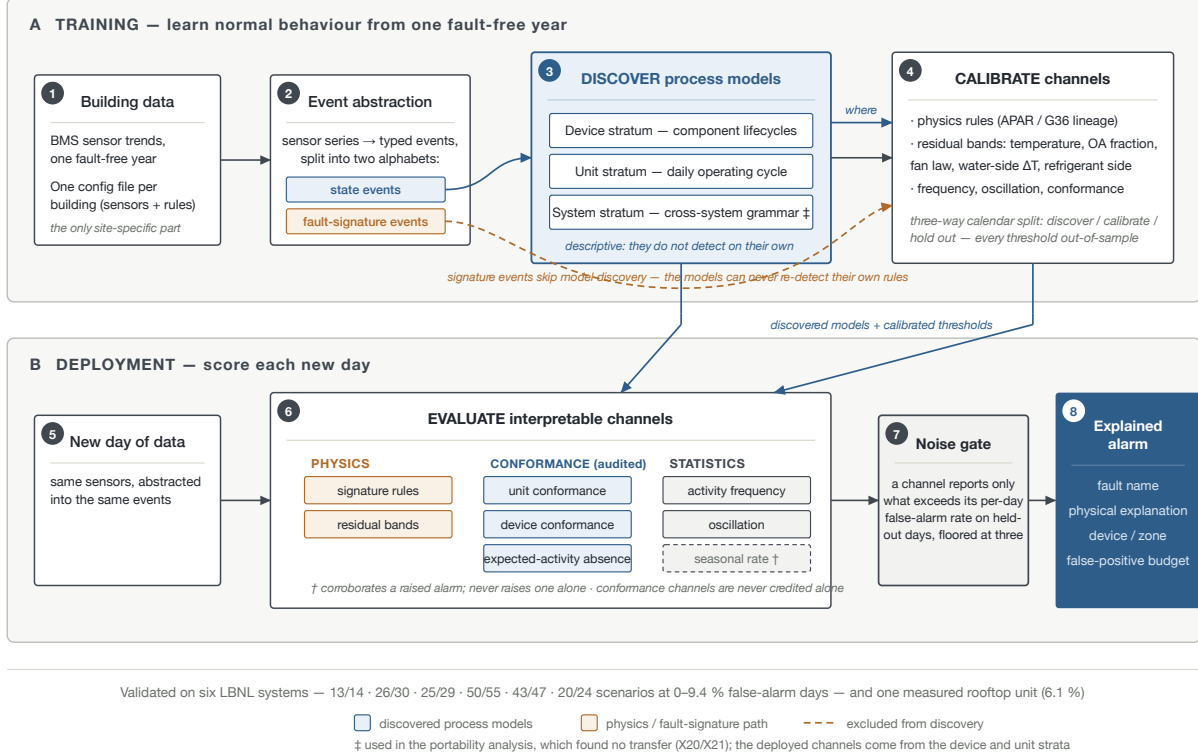


Figure 1: The framework as evaluated. Training (top): one fault-free year and one configuration file; events are split into state and fault-signature alphabets; process models are discovered from state events only and supply the strata and the invariants the channels use, but do not detect on their own; every threshold comes from a three-way calendar split, so every false-alarm rate is out-of-sample. Deployment (bottom): each day is scored by the physics, statistical and conformance channels; a channel reports only what exceeds its own measured noise floor; what survives reaches the operator with a name, a physical explanation, the device and the stated budget.

year only, split three ways by calendar: the last eight days of each month (96 days) are held out and report every false-alarm rate; the eight days before them form a calibration slice for the two conformance thresholds; the band channels learn their extremes on every other day and discovery uses the rest. No threshold sees fault data or the days its false-alarm rate is reported on. A channel then reports only what exceeds its own noise floor, a binomial test at $p < 10^{-3}$ against that channel’s holdout false-alarm rate floored at three days [26]; the rules channel, silent on the whole fault-free year by construction, is held to the same rule of three on the year. Detection thresholds are floored at sensor physics (0.5°F for temperature residuals), which retracted an early result of 21 to 117 detection-days per scenario to at most two. What survives the gate reaches the operator with the implicated device or zone, the evidence, and the channel’s false-alarm budget.

Table 1: The eight channels as deployed. Every threshold is set from the fault-free year alone. The rate channel is advisory: reported, never counted in the deployed union.

Channel	Reads	A day is flagged when	Threshold source
Rules	signature	a control-sequence rule fires during operation	documented bands; rules not silent on the fault-free year are removed at onboarding; gate 3 of 365
Residual	raw signals	a per-mode physical residual leaves its healthy band	[min, max] of training days, widened to a per-rule sensor floor
Model conformance	state	the day’s trace aligns to the unit net below threshold	1 % quantile of fitness on the calibration slice
Device conformance	state	an actuator’s lifecycle trace aligns to its device net below threshold	as above, per device net
Absence	state	a scheduled device produces no lifecycle event on a scheduled day	each device’s healthy silence rate
Frequency	state	a daily activity count leaves its healthy range	[min, max] over training days
Oscillation	raw signals	an actuator’s daily reversal count leaves its healthy range	[min, max] over training days
Rate (advisory)	state	a 30-day active-day rate leaves the range of month-matched healthy windows	training-year windows

5 The Falsification Protocol

An audit of 65 primary machine-learning HVAC fault-detection studies found two that shared code links, one of them broken, and 72% that did not say whether their data was public [42]. A field in that condition needs results that can be checked more than another accuracy figure, so verification was treated as a deliverable. The protocol has three stages and one ledger.

Before a result. Every experiment is pre-registered: the hypothesis and the observation that would refute it are committed to version control before the run [27, 46–48]. A falsifier that fires is a finding and is published. Ten results went against the project (four refuted hypotheses, one fault-informed rule, three withdrawn claims, one test without attainable power, one transfer test whose pass a discovery-free control reproduced) and four later tests fired their falsifiers; Appendix ?? lists every experiment with its prediction and outcome. Before any result is computed, a gate battery

checks the raw data: file hashes, monotonic timestamps, calendar consistency and rotation, Brick coverage [4], and event-log hashes against duplicate scenarios; a healthy-silence gate confirms per configuration that the rules are silent on the fault-free year. Three of the eight dataset errata were found by the gates, the rest by audit, and every one would have silently altered a published result.

Before a number is quoted. An adversarial audit recomputes the claimed quantity independently from raw data, by an auditor briefed to find it wrong (here, AI coding agents, denied the answer key for the sealed test). It caught a baseline detecting file provenance rather than faults; one single-duct detection that was a dataset artefact, so the score is 14 of 14 naive and 13 of 14 adjudicated; a significance test with an impossible null, retracted; detections below sensor precision; and, twice, a completed comparison against commercial practice, the first execution invalid for three reasons and the repaired one for six adapter defects and one defect of design (Section ??). None would have been caught by a test suite, because the code was doing what it was written to do.

After a result. About two hundred tests run on every commit; a regeneration gate re-derives every artefact and diffs it against the stored copy, so a changed value is a defect or a documented decision, never a silent update; every validated dataset is frozen as a fixture. The suite pins published values, fired falsifiers and disclosure obligations: a regeneration that drops a caveat fails the build exactly as a lost feature would. Continuous integration has guarded that a result regenerates [30]; we have not found it reported that it guards that a caveat survives.

What it taught. Sixty-seven numbered lessons carry three patterns: a lesson is not closed until a test enforces it, because documentation decays; every new channel recreated the same missing noise gate until the harness enforced it structurally; and simulation flatters, which is why the sensor-precision floor and the noise-floor requirement exist and why the one measured building is reported as a false-alarm rate, not a detection result. The datasets are public by DOI [21–23], the environment is pinned, seeds are fixed, and a claim-to-artefact table maps every published claim to the artefact that backs it.

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